



Unit 5

Ethical Decision in Software Development and Ethics of IT organization



Software Quality

- It refers to the degree to which a software product or system meets specified requirements and satisfies the needs or expectations of its users.
- It encompasses various attributes, characteristics, and factors that contribute to the overall excellence of a software application



Software Quality Importance

- **User Satisfaction:**High-quality software meets user expectations and provides a positive experience. Satisfied users are more likely to continue using the software and recommend it to others.
- **Customer Retention:**Quality software contributes to customer loyalty. Users are more likely to stick with a product that consistently delivers reliable and efficient performance.
- **Brand Reputation:**The quality of software directly affects the reputation of the company or product. Positive brand perception is crucial for attracting new customers and retaining existing ones.



- **Reduced Support and Maintenance Costs:** Well-designed and thoroughly tested software tends to have fewer defects, leading to lower support and maintenance costs. Fixing issues early in the development process is more cost-effective than addressing them post-deployment.
- **Competitive Advantage:** High-quality software sets a company apart from competitors. Users are more likely to choose a product that is known for its reliability and performance over alternatives with a history of quality issues.
- **Risk Mitigation:** Quality assurance practices help identify and address potential risks, such as security vulnerabilities, before they become major problems. This reduces the likelihood of data breaches, system failures, and other critical issues.



Strategy for developing Software Quality

- Define Clear Requirements: Work closely with stakeholders to gather and document clear and detailed requirements. Ensure that requirements are unambiguous, measurable, and testable.
- Plan Thoroughly: Develop a comprehensive project plan that includes timelines, milestones, and resource allocation. Consider risk management and contingency plans for potential issues.
- Use an Iterative Development Process: Adopt an iterative and incremental development methodology (e.g., Agile, Scrum). Regularly review and reassess project priorities and goals.



- **Automated Testing:** Implement a robust automated testing strategy, including unit tests, integration tests, and end-to-end tests. Use continuous integration tools to run tests automatically with each code commit.
- **Version Control:** Use version control systems (e.g., Git) to track changes and manage collaborative development. Tag releases and maintain a clear versioning system.
- **Performance Testing:** Conduct performance testing to ensure the software meets scalability and responsiveness requirements. Identify and address bottlenecks or performance issues.
- **Security Measures:** Implement secure coding practices. Regularly conduct security audits and vulnerability assessments.
- **User Feedback:** Collect and incorporate user feedback throughout the development process. Use beta testing or early releases to gather real-world feedback.
- **Continuous Improvement:** Conduct post-mortems or retrospectives after each release to identify areas for improvement. Encourage a culture of continuous learning and improvement within the development team.



Contingent Workers

- It also known as gig workers, freelancers, temporary employees, or independent contractors, are individuals hired by an organization on a non-permanent basis to fulfill specific roles or complete specific projects.
- Unlike full-time employees, contingent workers are not considered permanent staff members and are typically engaged for a defined period or task



Some Types of Contingent Workers:

- **Freelancers/Independent Contractors:** Individuals who are self-employed and provide services on a project-by-project basis.
- **Temporary Employees:** Hired for a specific period to fill temporary staffing needs.
- **Consultants:** Individuals with specialized skills or expertise brought in to provide advice or services for a particular project.
- **Part-time or Seasonal Workers:** Hired to work during specific times or seasons when additional workforce is needed.



Advantages of Contingent Workers:

- **Flexibility:**Contingent workers provide organizations with the flexibility to scale their workforce up or down based on project demands or seasonal fluctuations. This adaptability is particularly valuable in industries with variable workloads.
- **Cost Savings:**Hiring contingent workers can be cost-effective as organizations can avoid long-term financial commitments associated with full-time employees, such as benefits, taxes, and other overhead costs.
- **Access to Specialized Skills:**Contingent workers often bring specific expertise and skills that may not be readily available within the organization. Employing these individuals allows companies to access a diverse talent pool for specific projects or tasks.



- **Project-Based Work:**Contingent workers are well-suited for project-based work where a specific task has a defined timeline and scope. Once the project is complete, organizations can disband the team without the obligation of retaining full-time staff.
- **Quick Response to Market Changes:**In rapidly changing industries or markets, contingent workers enable organizations to quickly adapt to new trends or technologies by bringing in individuals with the necessary skills on a temporary basis.
- **Global Talent Pool:**Contingent workers can be sourced globally, allowing companies to access talent from different geographic locations. This helps foster diversity and provides access to a broader range of skills and perspectives.



H-1B Workers:

- They are a temporary(non-immigrant) worker that are employed in the US in a specialty occupation
- These specialty occupations typically require a higher education degree or its equivalent.
- Jobs in fields such as mathematics, engineering , technology and medical sciences often qualify



Some key points about H-1B workers:

- **Eligibility:** To qualify for an H-1B visa, individuals must have a job offer from a U.S. employer for a position that requires specialized knowledge and a minimum of a bachelor's degree or its equivalent.
- **Employer Sponsorship:** The H-1B visa is employer-sponsored, meaning that a U.S. employer must petition for the visa on behalf of the foreign worker. Employers are required to demonstrate that they are unable to find qualified US workers for the positions they are seeking to fill.
- **Duration:** H-1B visas are typically issued for an initial period of up to three years, with the possibility of extension for an additional three years. In certain circumstances, the total period of stay can be extended beyond six years.



- Quota: There is an annual numerical cap (quota) on the number of new H-1B visas that can be issued.
- Transition to Permanent Residency: Many H-1B workers pursue permanent residency (green card) in the United States. The employer may sponsor the worker for a green card, allowing them to become a permanent resident.
- Dependents: H-1B visa holders can bring their spouses and unmarried children under the age of 21 to the U.S. on H-4 visas



Outsourcing

- It is a business practice in which a company contracts out certain tasks or functions to external service providers rather than handling them internally.
- This can involve a variety of activities, ranging from manufacturing and information technology to customer support and human resources.
- It is often pursued for various reasons, including cost savings, access to specialized skills, and increased focus on core business functions.



Types of Outsourcing:

- Business Process Outsourcing (BPO): Involves contracting out specific business processes, such as customer service, human resources, or accounting.
- Information Technology Outsourcing (ITO): Involves outsourcing IT functions, including software development, maintenance, and support.



Reasons for Outsourcing:

- **Cost Savings:** One of the primary motivations for outsourcing is often the potential to reduce operational costs by leveraging the economies of scale and lower labor costs in other regions.
- **Focus on Core Competencies:** Outsourcing non-core functions allows a company to concentrate on its core business activities and strategic objectives.
- **Access to Specialized Skills:** Outsourcing can provide access to specialized skills and expertise that may not be available internally.



Challenges of Outsourcing:

- Communication Issues: Differences in language, culture, and time zones can lead to communication challenges.
- Quality Concerns: Maintaining quality standards can be a concern when outsourcing tasks to external providers.
- Security Risks: Outsourcing may involve sharing sensitive data with third parties, raising concerns about data security and privacy.
- Increase dependency
- Negative impact on staff



Whistle Blowing

- It refers to the act of exposing or disclosing information about wrongdoing, unethical behavior, or illegal activities within an organization or government.
- The person who raises concerns about such activities is often referred to as a whistleblower.
- Whistleblowers play a crucial role in promoting transparency, accountability, and ethical conduct within institutions.



Types

- Internal Whistleblowing: It involves disclosing information about misconduct, unethical behavior, or illegal activities within the organization they work for
- External Whistleblowing: It involves disclosing information about wrongdoing to parties outside the organization. This can include government agencies, regulatory bodies, law enforcement, the media, or advocacy groups



Dealing with a whistle blowing situation:

- **Establish Clear Reporting Mechanisms:** Ensure that the organization has well-defined and easily accessible channels for employees to report concerns internally. This may include hot lines, email, or other confidential reporting systems.
- **Promote a Culture of Openness:** Foster a workplace culture that encourages open communication and ethical behavior. Make it clear that the organization values integrity and will not tolerate retaliation against whistle blowers.
- **Investigate Promptly and Fairly:** Take all reports seriously and conduct thorough and impartial investigations. Assign the responsibility to an independent party if necessary to ensure objectivity.



- **Communicate Transparently:**Keep employees informed about the progress of the investigation without compromising confidentiality. Transparent communication helps demonstrate the organization's commitment to addressing concerns.
- **Corrective Action:**If the investigation reveals wrongdoing, take appropriate corrective action. This may include disciplinary measures, policy changes, or other steps to address and rectify the issues raised by the whistleblower.
- **Legal Compliance:**Ensure that the organization is compliant with whistleblower protection laws. Familiarize yourself with the specific legal requirements in your jurisdiction and implement policies accordingly.
- **Document the Process:**Keep detailed records of the entire whistleblowing process, including the receipt of the report, the steps taken during the investigation, and any actions or changes implemented as a result. This documentation can be important in the event of legal challenges.



Benefits associated with blowing the whistle:

- **Exposing Wrongdoing:**Whistleblowing allows individuals to bring attention to illegal or unethical activities within an organization. This exposure can lead to the investigation and rectification of the identified issues.
- **Promoting Accountability:**Whistleblowing holds individuals and organizations accountable for their actions. By disclosing misconduct, whistle blowers contribute to a culture of responsibility and integrity.
- **Preventing Harm:**Whistleblowers may reveal information that can prevent harm to individuals, the environment, or the public. This can include issues related to safety violations, environmental hazards, or other risks.



- **Protecting the Public Interest:**Whistleblowing can serve the greater good by bringing attention to matters that are of significant public interest. This may include cases of fraud, corruption, or activities that impact the welfare of the broader community.
- **Legal Compliance:**Whistleblowing can assist in ensuring that organizations comply with laws, regulations, and industry standards. Reporting illegal activities or non-compliance can prompt investigations and legal actions.
- **Catalyzing Change:**Whistleblowers may be catalysts for positive change within an organization. Their disclosures can lead to policy reforms, improved internal controls, and a stronger commitment to ethical practices.
- **Protecting Others:**Whistleblowers may take action to protect colleagues, customers, or other stakeholders who may be unaware of the wrongdoing. By speaking up, they contribute to a safer and more ethical work environment.



Green Computing

- It also known as sustainable or Eco-friendly computing, refers to the practice of designing, manufacturing, using, and disposing of computers and other electronic devices in an environmentally friendly manner
- Its goal is to minimize the impact of information technology on the environment by promoting energy efficiency, reducing electronic waste, and using environmentally sustainable materials.
- It is a multidisciplinary field that involves collaboration between computer scientists, environmental scientists, policymakers, and industry stakeholders.



Some key benefits of green computing:

- **Energy Efficiency:** Implementing energy-efficient practices and technologies leads to reduced electricity consumption, resulting in lower energy bills for individuals and organizations.
- **Reduced Carbon Footprint:** By optimizing energy usage and relying on renewable energy sources, green computing helps reduce the carbon footprint associated with information technology, mitigating climate change effects.
- **Extended Equipment Lifespan:** Designing devices with energy-efficient components and materials that are less prone to obsolescence can lead to longer-lasting products, reducing electronic waste.



- **Resource Conservation:** Green computing emphasizes the use of environmentally friendly materials and promotes recycling, reducing the need for raw materials and minimizing the impact on ecosystems.
- **E-Waste Reduction:** Implementing proper e-waste disposal and recycling programs helps prevent hazardous electronic waste from entering landfills and contributes to a circular economy.
- **Sustainable Supply Chain:** Working with suppliers who adhere to environmentally sustainable practices contributes to a greener supply chain.

